Exact methods

Branch & Bound for integer optimization

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Optimization: principles and algorithms

ÉCOLE POLYTECHNIQUE
FÉDÉRALE DE LAUSANNE

Branch & Bound for integer optimization

We illustrate the idea of the B&B method in the context of integer optimization.

Example

Problem P

$$\min x_1 - 2x_2$$

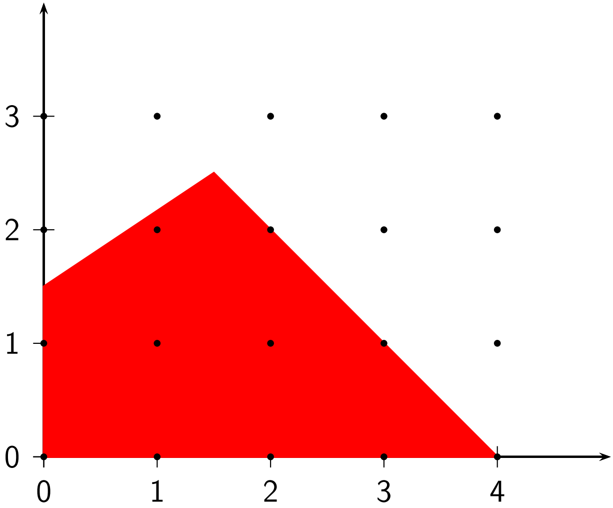
subject to

$$\begin{aligned} -4x_1 + 6x_2 &\leq 9 \\ x_1 + x_2 &\leq 4 \\ x_1, x_2 &\geq 0 \\ x_1, x_2 &\in \mathbb{N}. \end{aligned}$$

Upper bound

(0,0) is feasible: $U{=}0$

Example



Lower bound

Solve the relaxation. Erase the integrality constraints and call the problem $R(P)$.

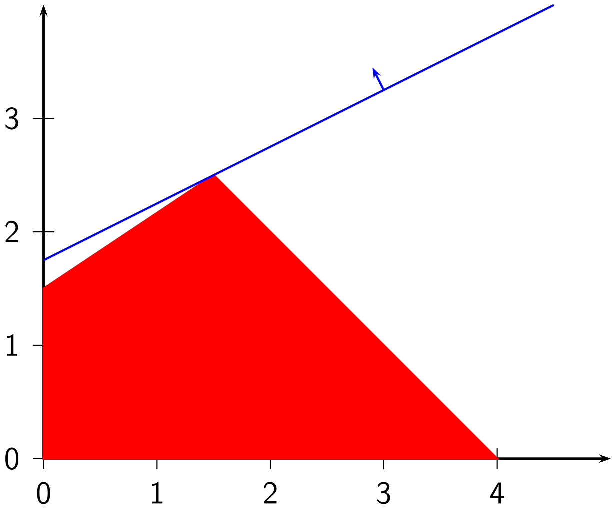
Lower bound

$$\min x_1 - 2x_2$$

subject to

$$\begin{aligned} -4x_1 + 6x_2 &\leq 9 \\ x_1 + x_2 &\leq 4 \\ x_1, x_2 &\geq 0 \\ x_1, x_2 &\in \mathbb{N}. \end{aligned}$$

Lower bound

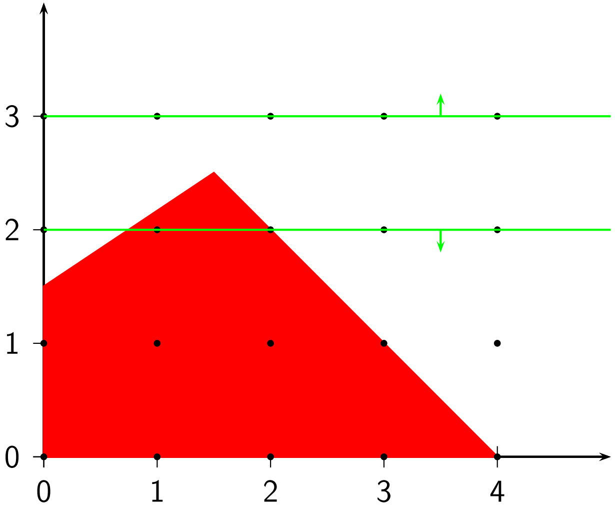


Divide

- Optimal solution of $R(P)$: $(1.5, 2.5)$
- $\ell(P)$: -3.5

P_1	P_2
$\min x_1 - 2x_2$	$\min x_1 - 2x_2$
s.c.	s.c.
$-4x_1 + 6x_2 \leq 9$	$-4x_1 + 6x_2 \leq 9$
$x_1 + x_2 \leq 4$	$x_1 + x_2 \leq 4$
$x_1, x_2 \geq 0$	$x_1, x_2 \geq 0$
$x_1, x_2 \in \mathbb{N}$	$x_1, x_2 \in \mathbb{N}$
$x_2 \leq 2$	$x_2 \geq 3$

Divide

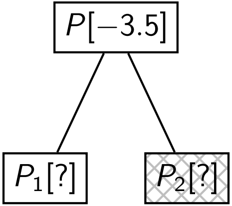


Tree representation

Upper bound

$U = 0$

Tree

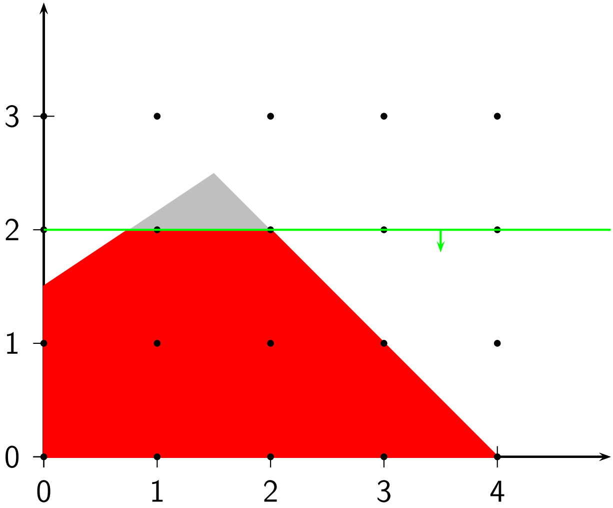


P_2 is infeasible.

P_1

Mention that the feasible set of P_1 is exactly the same as P.

Problem P_1



P_1: lower bound

Write the relaxed problem by hand.

P_1: lower bound

Problem P_1

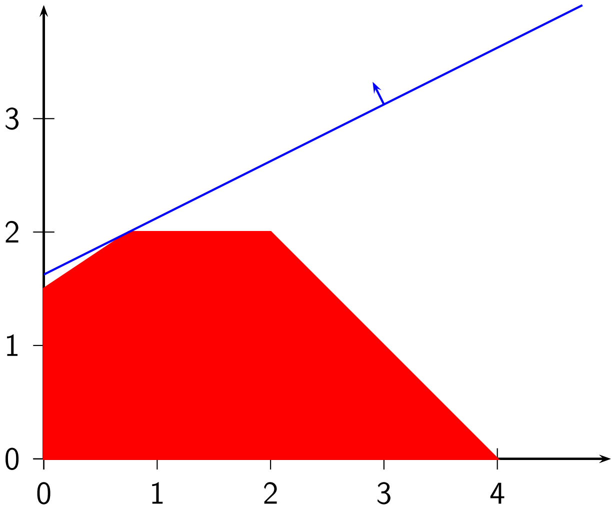
$$\min x_1 - 2x_2$$

subject to

$$\begin{aligned} -4x_1 + 6x_2 &\leq 9 \\ x_1 + x_2 &\leq 4 \\ x_1, x_2 &\geq 0 \\ x_2 &\leq 2 \\ x_1, x_2 &\in \mathbb{N}. \end{aligned}$$

Relaxation $R(P_1)$

P_1: lower bound

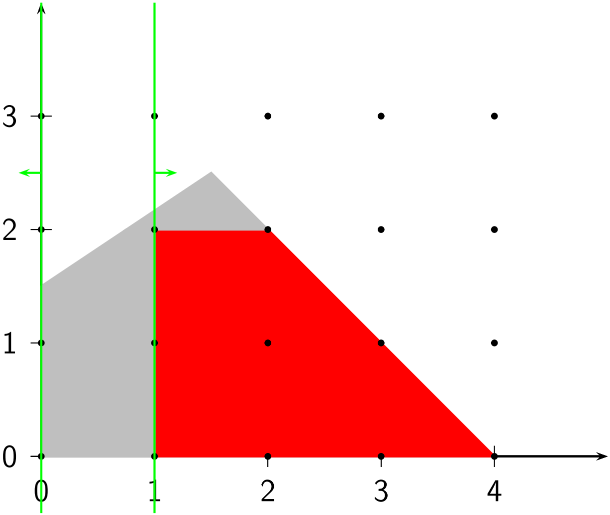


P_1: divide

- Optimal solution of $R(P_1)$: $(0.75, 2)$
- $\ell(P_1)$: -3.25

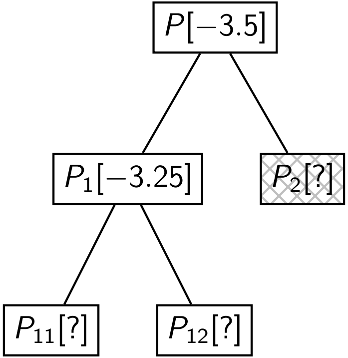
P_{11}	P_{12}
$\min x_1 - 2x_2$	$\min x_1 - 2x_2$
s.c.	s.c.
$-4x_1 + 6x_2 \leq 9$	$-4x_1 + 6x_2 \leq 9$
$x_1 + x_2 \leq 4$	$x_1 + x_2 \leq 4$
$x_1, x_2 \geq 0$	$x_1, x_2 \geq 0$
$x_1, x_2 \in \mathbb{N}$	$x_1, x_2 \in \mathbb{N}$
$x_2 \leq 2$	$x_2 \leq 2$
$x_1 \leq 0$	$x_1 \geq 1$

P_1: divide



Tree representation

$U = 0$



P_{11}: lower bound

Write the relaxed problem by hand.
Mention that $x_1 = 0$.

P_{11}: lower bound

Problem P_{11}

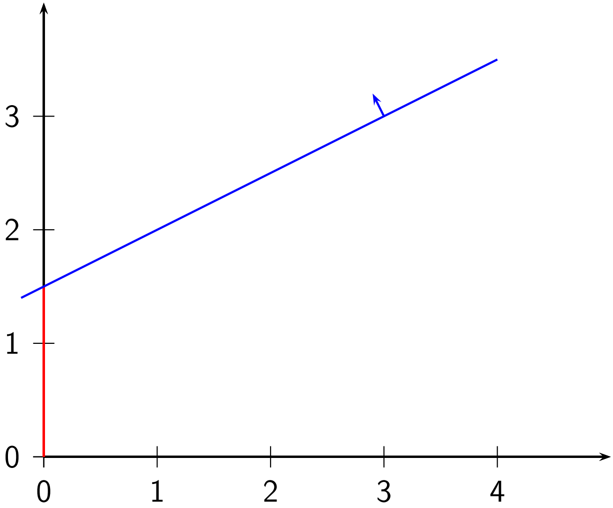
$$\min x_1 - 2x_2$$

subject to

$$\begin{aligned} -4x_1 + 6x_2 &\leq 9 \\ x_1 + x_2 &\leq 4 \\ x_1, x_2 &\geq 0 \\ x_2 &\leq 2 \\ x_1 &\leq 0 \\ x_1, x_2 &\in \mathbb{N}. \end{aligned}$$

Relaxation $R(P_{11})$

P_{11}: lower bound



P_{11}: lower bound

- Optimal solution of $R(P_{11})$: $(0, 1.5)$
- $\ell(P_{11})$: -3

P_{12}: lower bound

Write the relaxed problem by hand.

P_{12}: lower bound

Problem P_{12}

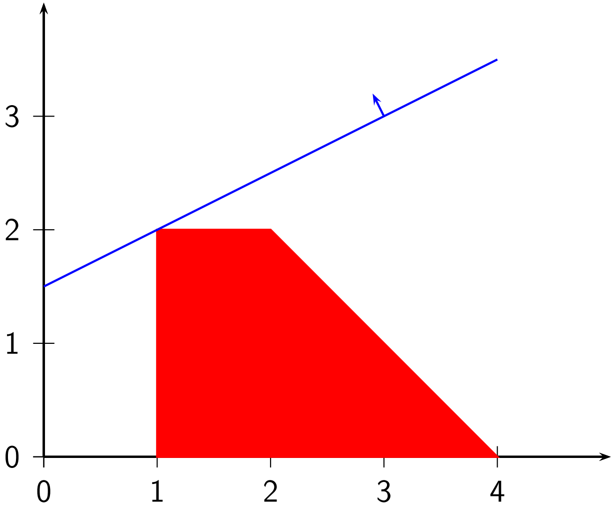
$$\min x_1 - 2x_2$$

subject to

$$\begin{aligned} -4x_1 + 6x_2 &\leq 9 \\ x_1 + x_2 &\leq 4 \\ x_1, x_2 &\geq 0 \\ x_2 &\leq 2 \\ x_1 &\geq 1 \\ x_1, x_2 &\in \mathbb{N}. \end{aligned}$$

Relaxation $R(P_{12})$

P_{12}: lower bound



P_{12}: lower bound

- Optimal solution of $R(P_{12})$: $(1, 2)$
- $\ell(P_{12})$: -3

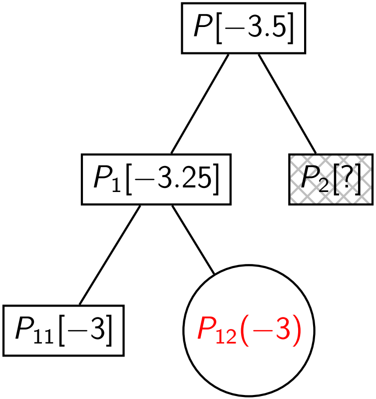
Integer solution
It is the optimal solution of P_{12}.
Feasible solution for P
Better upper bound: $U = -3$.

Tree

Clean the tree by removing subproblem P_1 because its lower bound is not better than the upper bound.
Then show that we have found any optimal solution

Tree representation

$U = -3$



Summary

The branch & bound method for integer optimization applies the divide and conquer concept progressively.
Branching: taking a fractional variable, and branch on it.
Lower Bound: solving the relaxation
Upper bound: best feasible solution found so far.